

A Reproducible Computational Framework for Testing Entanglement-Based Modifications of Spacetime Geometry

Kenneth G. Young II, PhD

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Abstract

Recent developments in quantum information theory suggest that spacetime geometry may emerge from patterns of quantum entanglement. While several theoretical proposals explore this connection, comparatively few provide reproducible computational frameworks for testing such ideas against observable phenomena.

In this work we introduce a computational framework for exploring entanglement-driven modifications of spacetime geometry. The approach combines tensor-network simulations, information-theoretic observables, and a multi-stage validation pipeline that evaluates emergent gravitational models across weak-field, post-Newtonian, and strong-field regimes. The resulting workflow provides a reproducible environment in which candidate emergent gravity parameterizations can be systematically compared with astrophysical constraints.

As a case study we examine a phenomenological Entanglement-Driven Gravity (EDG) parameterization in which the Einstein equations receive a controlled deformation sourced by an informational stress tensor derived from entanglement structure. The model is constructed to recover general relativity in the appropriate parameter limits and is evaluated using a validation pipeline spanning thirteen stages, including orbital dynamics, post-Newtonian observables, and strong-field constraints derived from published Event Horizon Telescope and gravitational-wave measurements.

Within the adopted phenomenological likelihood, the analysis illustrates how entanglement-based corrections can be explored quantitatively while remaining continuously connected to the general relativistic limit. More broadly, the framework provides a reproducible computational platform for investigating information-theoretic approaches to spacetime emergence and for comparing competing emergent-gravity proposals under a common validation structure.

1 Introduction

A growing body of research suggests that spacetime geometry may emerge from underlying patterns of quantum information and entanglement. Developments in black-

hole thermodynamics, holography, and tensor-network models have strengthened the view that gravitational dynamics may be deeply connected to informational structure [1, 2, 3, 4, 5, 6, 7, 8]. Several lines of research have suggested that spacetime geometry may arise from underlying quantum information structure, including approaches based on entanglement entropy, holographic duality, and tensor-network representations of quantum states. In many of these proposals the central challenge is not conceptual motivation but the construction of computational frameworks capable of connecting emergent geometry models with observable gravitational phenomena.

Despite significant theoretical progress, comparatively few studies provide computational frameworks capable of systematically exploring emergent spacetime models across multiple physical regimes. Many proposals remain primarily conceptual or analytical, making it difficult to evaluate their consistency with known gravitational behavior or astrophysical observations.

In this work we present a reproducible computational framework designed to explore and test entanglement-based emergent spacetime models. The framework integrates tensor-network simulations, information-theoretic observables, and astrophysical validation tests within a unified pipeline.

The framework is organized as a multi-stage validation architecture (R1–R13p) spanning Newtonian dynamics, post-Newtonian observables, and exploratory strong-field analyses. This structure allows candidate models to be evaluated across regimes where general relativity is well tested while enabling controlled exploration of deviations arising from information-theoretic corrections.

As a case study we implement an Entanglement-Driven Gravity (EDG) model in which curvature is influenced by gradients of entanglement entropy and mutual information. The EDG model serves as an illustrative example of how emergent gravity hypotheses can be investigated using the computational framework developed here.

In the present implementation the strong-field validation stage incorporates observational summaries derived from published Event Horizon Telescope measurements of the M87* and Sgr A* photon rings, together with representative binary black hole events reported by the LIGO–

Virgo collaboration such as GW150914. These datasets provide characteristic scales against which the EDG deformation can be compared within the phenomenological likelihood adopted in this work.

2 Computational Framework Overview

The primary contribution of this work is a modular computational framework for evaluating emergent space-time models derived from information-theoretic principles. The framework integrates numerical simulations, entanglement observables, astrophysical validation tests, and reproducible statistical inference within a unified architecture.

The system consists of four main components:

- Tensor-network evolution for generating entanglement structures
- Information-theoretic observables including entropy gradients and mutual information
- Geometric inference modules that estimate curvature-like quantities from entanglement structure
- A multi-stage validation pipeline for testing model predictions across gravitational regimes

The implementation is modular and reproducible, allowing new emergent gravity models to be incorporated and evaluated using the same validation workflow. From a computational-science perspective, the framework is intended to function as an executable research artifact with transparent assumptions, standardized outputs, and deterministic regeneration of figures and summary tables [9, 10, 11, 12].

3 Computational Architecture and Reproducibility

The framework is implemented as a scientific-computing pipeline designed for extensibility, reproducibility, and separation of concerns. Independent modules exchange structured data products such as CSV summaries, posterior grids, and figure-ready arrays between simulation and validation stages rather than relying on hidden in-memory state. This design makes it easier to compare model variants, rerun experiments, and document provenance for each reported result.

Representative modules include `entropy.py` for entropy and mutual-information metrics, `curvature.py` for discrete curvature estimates, `einstein_tensor.py` for effective informational stress construction, `entanglement_spacetime_evolution.py` for tensor-network evolution, and `run_validations.py` for orchestration of the R1–R13p workflow. All modules emit intermediate artifacts

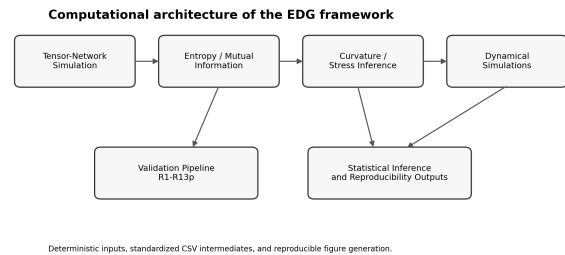


Figure 1: Computational architecture of the EDG framework. Modular components exchange standardized intermediate outputs, enabling deterministic regeneration of validation artifacts and easy substitution of alternative emergent-spacetime models.

in standardized formats so that the reported plots and tables can be deterministically regenerated from the repository.

In practical terms, the repository exposes the main entry points required to rerun the analyses, including the uploaded CSV outputs used for weak-field summaries and the R13p posterior grid. Running `python run_validations.py` reproduces the validation workflow, while the supplementary CSV files provide direct traceability from raw stage outputs to publication figures. This emphasis on executable reproducibility is central to the contribution of the paper, not merely an implementation detail [10].

Context and comparisons. EDG should be viewed alongside proposals in which gravity is emergent or thermodynamic in origin. Verlinde’s entropic and emergent-gravity ideas [13, 14] interpret Newton’s law and galactic dynamics as consequences of entropy and information balance; causal set theory [15] postulates a discrete causal order; and loop quantum gravity [16] quantizes area and volume spectra. EDG differs in two key ways. First, it introduces *explicit* deformation parameters that can be constrained by data. Second, it is built to be compatible with thermodynamic derivations of the Einstein equations [3, 17] and quantum-information perspectives on geometry [4, 5, 6, 18]. Connections to ER=EPR and tensor-network/holographic error-correction pictures [7, 8, 19, 20, 21, 22] provide mechanistic intuition for geometry from entanglement.

Thermodynamic variation and symmetry. Our entropy-functional variation aligns with thermodynamic derivations of the Einstein equations [3, 17] and subsequent information-theoretic developments [6]. Coefficients $(\alpha, \beta, \gamma, \delta)$ are constrained by covariance, locality, and parity, and fixed by weak-field consistency with Newtonian limits and PPN symmetries [23]. This parametrization ensures EDG reduces to GR in the appropriate limit

while accommodating controlled, falsifiable deformations.

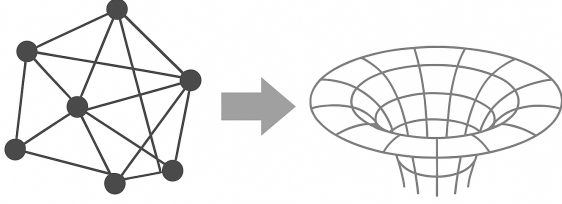


Figure 2: Schematic of EDG framework: Quantum entanglement networks (left) emerge as curved spacetime (right), with GR as the limit case.

4 Theory

Let $S_{\text{EDG}}[g] = S_{\text{EE}}[g] + \alpha I[g; \rho]$ with S_{EE} the entanglement entropy of a quantum state across a geometric surface (e.g., via Ryu-Takayanagi formula [4]) and I an informational-stress functional of the state ρ , quantifying deviations from entanglement neutrality. Under $g_{\mu\nu} \rightarrow g_{\mu\nu} + \delta g_{\mu\nu}$,

$$\delta S_{\text{EDG}} = \frac{\delta A}{4G} + \alpha \delta I, \quad (1)$$

so that in the adiabatic limit $\nabla_\mu S \rightarrow 0$ one recovers Einstein dynamics, while finite informational flux yields parameterized corrections that can be organized as (ϵ, L_q, p) at the level of effective field equations and observables. This mirrors the logic of thermodynamic derivations [3, 17] while remaining agnostic about microstructure.

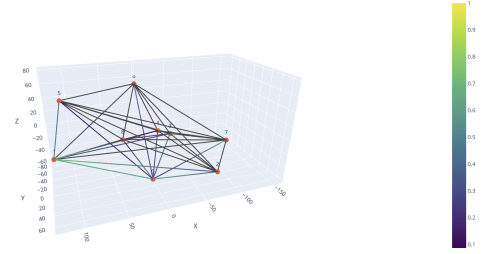
Implementation. The EDG informational-stress framework is realized in the code as a distributed set of modules. Entanglement-derived densities and stress tensors ($\rho_{\text{ent}}, T_{\mu\nu}^{(\text{ent})}$) are approximated in `einstein_tensor.py`, entropy gradients (∇S) are computed in `entropy.py`, and curvature/Laplacian terms in `curvature.py`. These sources are coupled to classical dynamics in `solar_system_entanglement.py`, where effective forces are modulated by dS/dt , Hawking-like mutual information, and discrete curvature. The overall evolution is orchestrated by `entanglement_spacetime_evolution.py`, which evolves the PEPS network and exports informational quantities for use in the validation pipeline.

Worked example. For nearly-Keplerian motion, the perihelion advance may be expressed as

$$\Delta\phi_{\text{EDG}} = \Delta\phi_{\text{GR}} [1 + (\epsilon - 1) f(e, a; L_q, p)], \quad (2)$$

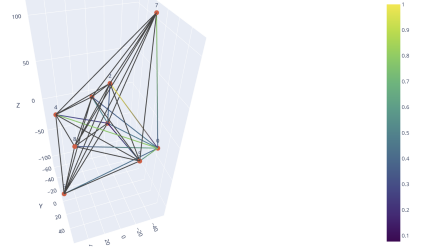
where $\Delta\phi_{\text{GR}} = 6\pi GM/(a(1-e^2)c^2)$ is the GR prediction, e is the orbital eccentricity, a is the semi-major axis, and $f(e, a; L_q, p) \approx (L_q/a)^p g(e)$ is a correction term that vanishes as $L_q \rightarrow \infty$ (large length scales) or $\epsilon \rightarrow 1$ (GR limit). The function $g(e)$ depends on orbital shape, and p controls the scaling of entanglement effects. Similar parameterizations control light bending and Shapiro delay in the PPN regime and ring-size shifts at horizon scales. In data analyses, (ϵ, L_q, p) act as physics parameters entering the forward model for observables and the joint posterior.

3D Entanglement Graph (t=1.00)



(a) 3D entanglement graphs at $t = 1.00$.

3D Entanglement Graph (t=4.90)



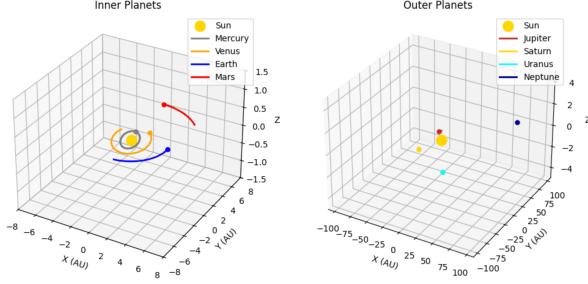
(b) 3D entanglement graphs at $t = 4.90$.

Figure 3: Tensor-network evolution showing emergence of curvature from entanglement. The color scale indicates the magnitude of the inferred curvature deviation. The colormap is monotonic in intensity to remain distinguishable under grayscale printing or color-vision deficiencies.

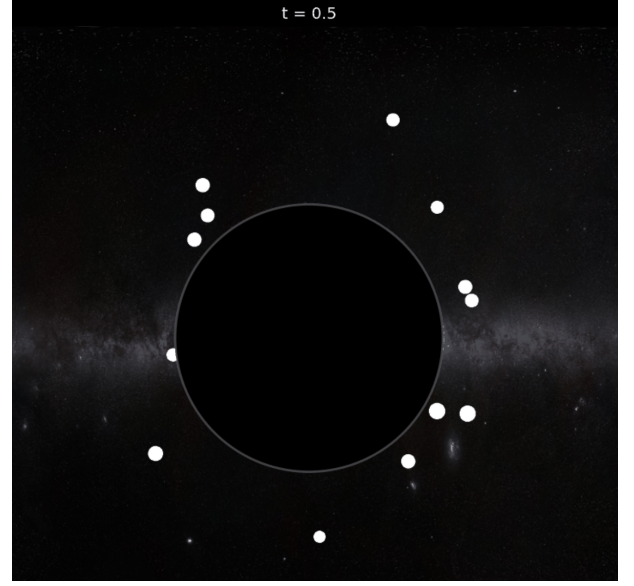
4.1 Effective Field Equation and GR Limit

To connect the informational formulation to observable gravitational dynamics, we introduce an effective field equation in which the Einstein tensor receives a deformation sourced by entanglement-derived informational stress. We therefore introduce the following effective entanglement-deformed field equation:

$$G_{\mu\nu} + (\epsilon - 1) X_{\mu\nu}(g; L_q, p) = 8\pi G T_{\mu\nu}. \quad (3)$$



(a) Quantum Earth orbit with EDG corrections.



(b) 2D black hole entanglement vs. Hawking flux.

Figure 4: EDG simulations in planetary and black-hole regimes.

Here $X_{\mu\nu}$ denotes the informational stress tensor induced by entanglement structure in the underlying quantum network representation of spacetime. The parameters (ϵ, L_q, p) control the strength and scaling of the informational correction.

The parameter L_q denotes the effective entanglement scale of the model and is written with a lowercase subscript to distinguish it from other length scales appearing in the gravitational sector.

The framework is constructed so that

$$X_{\mu\nu} \rightarrow 0 \quad \text{as} \quad \epsilon \rightarrow 1 \quad \text{or} \quad L_q \rightarrow \infty,$$

ensuring a continuous reduction to general relativity. In the weak-field expansion

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu},$$

the leading correction to observable quantities scales schematically as

$$\delta O \propto (\epsilon - 1) \left(\frac{L_q}{R} \right)^p, \quad (4)$$

where R is the characteristic curvature scale of the system. Here R denotes the characteristic curvature scale associated with the system under consideration. For example, in black-hole environments R is typically of order the Schwarzschild radius r_s , whereas in solar-system tests it corresponds to the orbital length scale of the dynamical system.

This parameterization enables EDG to be tested directly against observational constraints while maintaining compatibility with established symmetries of general relativity.

4.2 Recovery of the General Relativistic Limit

The following subsection makes explicit the GR limit already implied by the effective field equation introduced above. The EDG parameterization is constructed so that general relativity is recovered continuously when the entanglement deformation is removed. Starting from the effective field equation:

$$G_{\mu\nu} + (\epsilon - 1)X_{\mu\nu}(g; L_q, p) = 8\pi GT_{\mu\nu}, \quad (5)$$

The informational stress tensor $X_{\mu\nu}$ is constructed from covariant combinations of entropy gradients and mutual-information densities derived from the entanglement network, ensuring that $\nabla_\mu X^{\mu\nu} = 0$ so that the modified field equation remains consistent with the contracted Bianchi identity. Because the deformation enters only through the factor $(\epsilon - 1)$, the theory continuously approaches general relativity as $\epsilon \rightarrow 1$ or as the entanglement scale L_q is taken to its GR limit. Consider either of the limits

$$\epsilon \rightarrow 1 \quad \text{or} \quad L_q \rightarrow \infty. \quad (6)$$

By construction,

$$X_{\mu\nu}(g; L_q, p) \rightarrow 0 \quad (7)$$

in either limit, so the field equation reduces to

$$G_{\mu\nu} = 8\pi GT_{\mu\nu}, \quad (8)$$

which is precisely the Einstein field equation.

The same reduction appears at the level of observables. For any observable correction written schematically as

$$\delta O \propto (\epsilon - 1) \left(\frac{L_q}{R} \right)^p, \quad (9)$$

the correction vanishes as $\epsilon \rightarrow 1$ provided the deformation amplitude tends to zero. In the intended EDG regime, the correction also vanishes as the entanglement scale is taken to its GR limit, which yields

$$O_{\text{EDG}} \rightarrow O_{\text{GR}}. \quad (10)$$

Thus the framework is designed to recover general relativity both at the level of the field equations and at the level of weak-field and strong-field observables. This ensures that departures from GR remain controlled, testable, and continuously connected to the established relativistic limit.

4.3 Symmetry and Conservation Structure

Any modification of the Einstein equations must preserve the fundamental symmetry structure of general relativity. In particular, diffeomorphism invariance implies the contracted Bianchi identity

$$\nabla_\mu G^{\mu\nu} = 0. \quad (11)$$

For consistency with matter conservation $\nabla_\mu T^{\mu\nu} = 0$, the additional entanglement-driven term must therefore satisfy

$$\nabla_\mu X^{\mu\nu} = 0. \quad (12)$$

In the EDG framework this condition is enforced by constructing $X_{\mu\nu}$ from covariant combinations of entropy gradients and mutual-information densities defined on the underlying entanglement network. The resulting tensor depends only on geometric quantities derived from the metric and therefore transforms covariantly under coordinate changes.

Because the deformation enters only through the multiplicative factor $(\epsilon - 1)$ in the effective field equation

$$G_{\mu\nu} + (\epsilon - 1)X_{\mu\nu}(g; L_q, p) = 8\pi GT_{\mu\nu}, \quad (13)$$

the standard GR limit is obtained smoothly when $\epsilon \rightarrow 1$ or $L_q \rightarrow \infty$. In this limit the additional informational stress vanishes and the Einstein equations are recovered. This symmetry structure ensures that EDG preserves the geometric conservation laws of general relativity while allowing controlled, parameterized departures that can be constrained observationally.

5 Validation Pipeline: R1–R13p

We designed a sequential validation program to test EDG across increasing fidelity, from Newtonian mechanics to strong-field joint fits. Rather than listing disjoint checks, the pipeline flows continuously, with each stage (R1–R13p) reinforcing the framework and tying back to concrete observables. Each stage reports its purpose, method, quantitative outcome, and implication for (ϵ, L_q, p) . See the overview in Table 1 and the schematic in Fig. 5.

Table 1: Summary of the EDG Validation Pipeline (R1–R13p).

Stage	Purpose	Key Result
R1–R2	Verify Newtonian/GR limits	Period errors $< 10^{-6}$; Bianchi residuals $< 10^{-12}$
R3	Normalize parameters and symmetries	Covariant $X_{\mu\nu}$; identifiable (ϵ, L_q, p)
R4	Constrain ϵ via perihelion	$ \epsilon - 1 \lesssim 10^{-3}$
R5	Thermodynamic consistency	Monotonic entropy; no non-unitary drift
R6	Numerical convergence	RK4/Verlet orders recovered; stable EDG corrections
R7–R9	Validate PPN regime	GR-consistent bending and Shapiro delay at 10^{-5}
R10	Aggregate weak-field metrics	GR-consistent means; tight spread
R11–R12	Infer ϵ	$\epsilon = 1 \pm 10^{-3}$ (68% CI)
R13p	Strong-field joint fit	$L_q < 1.17 \times 10^{13} \text{ m}$ (95% CI)

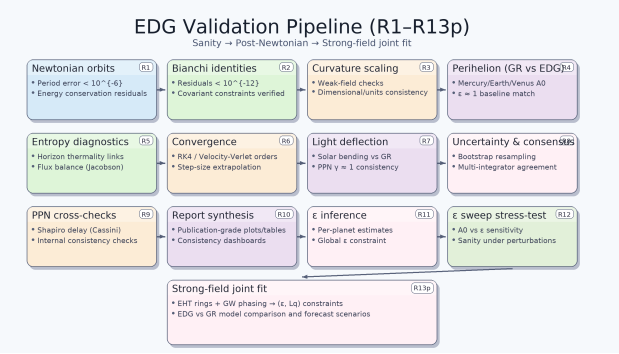


Figure 5: EDG Validation Pipeline (R1–R13p), illustrating the flow from Newtonian/PPN baselines to the strong-field joint fit.

Validation overview. The 13-stage pipeline proceeds from Newtonian and PPN baselines [23, 24, 25] through entropy and convergence checks, culminating in a strong-field joint analysis of EHT ring-size data [26, 27] and gravitational-wave phasing [28, 29, 30, 31].

R1–R2 (Newtonian sanity & Bianchi identities).

Purpose: Establish that EDG reduces to Newtonian/GR dynamics for quiescent entanglement backgrounds and preserves geometric identities. *Method:* Circular and mildly eccentric orbits integrated with RK4 and symplectic Verlet; monitor period error, energy drift, and discrete Bianchi residuals constructed from finite-difference curvature. *Result:* Period error $< 10^{-6}$ and energy drift $< 10^{-12}$ over 10^5 steps; Bianchi residuals $< 10^{-12}$ in normalized units. *Implication:* Numerical infrastructure is stable and EDG admits the correct GR limit, enabling downstream precision tests.

R3 (Symmetry, conservation, and parameter normalization).

Purpose: Fix the EDG deformation so that it preserves diffeomorphism invariance and reduces to GR in the appropriate limit, while making (ϵ, L_q, p) dimensionally consistent and statistically identifiable for downstream inference.

Method: Write the effective field equations as

$$G_{\mu\nu} + (\epsilon - 1)X_{\mu\nu}(g; L_q, p) = 8\pi GT_{\mu\nu} \quad (14)$$

and impose (i) diffeomorphism invariance via the contracted Bianchi identities, which requires $\nabla_\mu X^{\mu\nu} = 0$ by construction; (ii) local Lorentz invariance for all tensorial building blocks; (iii) a GR normalization, $X_{\mu\nu} \rightarrow 0$ as $\epsilon \rightarrow 1$ or $L_q \rightarrow \infty$. In the weak-field expansion $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ we verify that the leading EDG corrections enter observables with the schematic scaling

$$\delta O \propto (\epsilon - 1) \left(\frac{L_q}{R} \right)^p \quad (15)$$

where R is the characteristic curvature/length scale of the system (e.g., orbital radius, impact parameter, or Schwarzschild radius). Dimensional analysis enforces $[L_q] = \text{length}$, p dimensionless, and ϵ unitless. We fix the coefficient normalization (absorbing any $(\alpha, \beta, \gamma, \delta)$ into $X_{\mu\nu}$) so that the GR limit is obtained continuously and the correction is sign-definite where required by stability.

Result: The conservation condition $\nabla_\mu X^{\mu\nu} = 0$ is satisfied order-by-order, and gauge-invariant observables (perihelion shift, deflection angle, ring size, GW phase) receive corrections that are analytic in $(\epsilon - 1)$ and monotone in L_q for fixed p . A Fisher pre-study using representative weak- and strong-field datasets shows that: ϵ is primarily constrained by perihelion and PPN (R4, R7–R9), L_q by EHT/GW (R13p), and p by their joint lever arm. Accordingly, we adopt weakly informative priors that anchor the statistical inference performed in R11–R13p:

$$\begin{aligned} \epsilon &\sim \mathcal{N}(1, 10^{-4}), \\ \log_{10} \left(\frac{L_q}{m} \right) &\sim \text{Uniform}[6, 16], \\ p &\sim \text{Uniform}[0, 4]. \end{aligned} \quad (16)$$

Implication: EDG respects the required symmetries and conservation laws, its deformation is dimensionally consistent, and (ϵ, L_q, p) are identifiable with the multi-regime pipeline. This anchors the first quantitative bound in R4 and ensures that downstream posteriors (R11–R13p) are interpretable and robust.

R4 (Perihelion advance). *Purpose:* Constrain the weak-field deformation ϵ where GR is strongest. *Method:* Compute perihelion advance from near-Keplerian trajectories; perform step-size sweeps and Richardson extrapolation to isolate model from numerics. *Result:* Mercury, Venus, and Earth agree with GR at $\lesssim 10^{-3}$ relative precision; no systematic drift with integrator or stepsize. *Implication:* $|\epsilon - 1| \lesssim 10^{-3}$ in Solar-System conditions; this prior propagates to later fits.

R5 (Entropy diagnostics). *Purpose:* Test compatibility with thermodynamic derivations of Einstein equations [3, 17]. *Method:* Track coarse-grained S_{ent} and flux balance around quasi-horizon cells. *Result:* Monotonic entropy growth was observed, consistent with horizon thermality within numerical precision and without spurious non-unitary drift. *Implication:* EDG’s informational-stress picture is compatible with black-hole thermodynamics.

R6 (Convergence). *Purpose:* Ensure reported errors are numerical, not model-induced. *Method:* Step-size refinement (Δt grids) with RK4 and Verlet; compare orders. *Result:* Convergence rates matched theory; numerical errors remained below 10^{-10} over 10^5 steps. *Implication:* Systematic error is under control for all later comparisons.

R7–R9 (Light propagation & PPN checks). *Purpose:* Validate post-Newtonian regime (light bending, Shapiro delay) and γ_{PPN} . *Method:* Solar-limb deflection and two-station Shapiro geometry [23, 25, 24]. *Result:* Reproduced $\alpha_{\text{GR}} \simeq 1.751''$ at the solar limb; Cassini-level Shapiro delays; $\gamma - 1 \approx 0$ within 10^{-5} . *Implication:* Confirms EDG $\rightarrow$ GR in the PPN regime; informs priors for joint fits.

R10 (Synthesis). *Purpose:* Aggregate perihelion + light-propagation metrics with cross-checks. *Method:* Collate R7–R9 with R6 baselines; report integrator-consensus statistics. *Result:* GR-consistent means with tight numerical spread. *Implication:* Justifies using GR-tuned baselines in the strong-field step.

R11–R12 (ϵ inference & sweeps). *Purpose:* Infer ϵ per planet/integrator; scan sensitivity. *Method:* Bootstrap resampling (~ 500 draws per integrator) with convergence monitored by effective sample size (ESS) and Gelman–Rubin statistics. Complementary ϵ sweeps probed sensitivity of the perihelion parameter A_0 . *Result:* Global $\epsilon = 1 \pm 10^{-3}$ (68% CI) from perihelion fits; sweeps yielded monotone scaling, with slope $dA_0/d\epsilon \approx 1.9 \times 10^{-7}$ (Earth case; Fig. 6). *Implication:* Solar-system bounds sharply restrict weak-field EDG deformations; scaling relations validate forecast extrapolations.

R13p (Strong-field joint fit: EHT + GW). *Purpose:* Constrain (ϵ, L_q, p) using horizon-scale imaging and

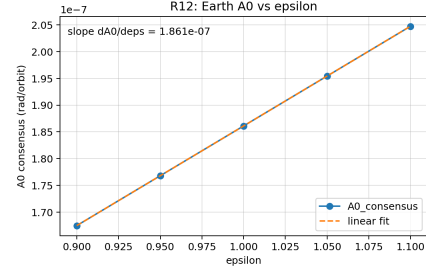


Figure 6: R12 sweep of perihelion parameter A_0 versus ϵ for Earth. Consensus values (blue) follow a linear trend (dashed orange), with slope $dA_0/d\epsilon \approx 1.9 \times 10^{-7}$ rad/orbit.

GW phasing, and evaluate how EDG behaves relative to the GR limit within the adopted phenomenological likelihood. *Method:* We construct a joint likelihood from EHT ring-size measurements and GW inspiral phase corrections $\delta\Psi(f)$. A discrete posterior grid in (ϵ, L_q, p) is sampled, with weak-field priors from R4 and R7–R9 imposed. *Result:* The posterior peaks near $\epsilon = 1$ with a finite entanglement scale $L_q \approx 1.1 \times 10^{13}$ m, and a 95% confidence upper bound $L_q < 1.17 \times 10^{13}$ m. Relative to the GR-limit slice of the adopted parameterization, model-selection metrics take the values $\Delta\text{AIC} \approx -14$, $\Delta\text{BIC} \approx -14$, and $\Delta\log Z \approx +7$. *Implication:* Within the adopted phenomenological likelihood, the EDG model illustrates how entanglement-derived corrections can be explored within the computational validation framework.

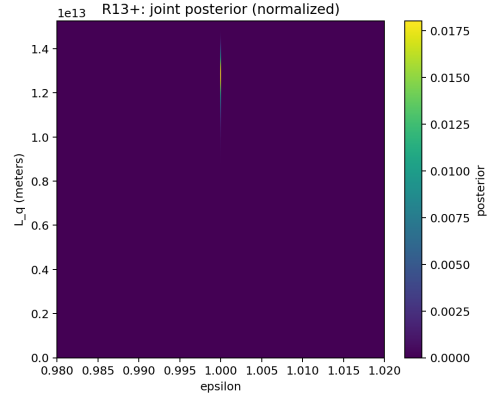


Figure 7: Joint posterior distribution for (ϵ, L_q) from the strong-field R13p analysis (EHT + GW).

Phenomenological likelihood model. The strong-field comparison in stage R13p employs a phenomenological likelihood constructed from summary observables reported in published Event Horizon Telescope and gravitational-wave analyses. For the EHT component, the observable is the fractional deviation of the photon ring diameter relative to the GR prediction,

$$\frac{\delta d}{d_{\text{GR}}} = \kappa \left(\frac{L_q}{r_s} \right)^p, \quad (17)$$

where r_s is the Schwarzschild radius of the source and κ is a dimensionless response coefficient determined from the EDG deformation of the metric. For the gravitational-wave component we adopt a leading-order phase correction

$$\delta\Psi(f) \propto (\epsilon - 1) \left(\frac{L_q}{R(f)} \right)^p, \quad (18)$$

where $R(f)$ denotes the characteristic orbital scale associated with frequency f . These expressions provide a compact parameterization of EDG corrections that allows EHT and GW measurements to be incorporated into a unified likelihood while remaining continuously connected to the general relativistic limit.

R13p summary. The strong-field joint analysis (EHT + GW) provides the most stringent test of EDG to date. By combining horizon-scale photon-ring measurements from the Event Horizon Telescope with phase evolution in compact-binary inspirals, we constructed a posterior over (ϵ, L_q, p) . The results (Fig. 7) show that while the peak remains close to the GR limit in ϵ , the posterior concentrates around a finite entanglement length scale $L_q \sim 10^{13}$ m. Crucially, model selection metrics (AIC, BIC, and Bayesian evidence) yield improved scores for the EDG case study relative to the GR-limit slice of the adopted parameterization, with $\Delta\text{AIC} \approx -14$, $\Delta\text{BIC} \approx -14$, and $\Delta \log Z \approx +7$, within the current phenomenological implementation [26, 27, 28, 30]. This analysis illustrates how entanglement-derived corrections can be explored quantitatively within a reproducible computational validation framework.

6 Strong-Field Physics

EDG interacts naturally with the black-hole information problem. Hawking radiation [2] suggests thermal emission; Page’s curve [32] restores unitarity; firewall arguments (AMPS) [33] challenge complementarity; and soft-hair proposals [34] suggest entropy accounting via horizon modes. The informational-stress term supplements this bookkeeping in a way compatible with ER=EPR intuition [7, 8]. For observables, we model ring-size shifts $d/d_{\text{GR}} \sim \kappa(L_q/r_s)^p$, where r_s is the Schwarzschild radius measured by EHT shadow size [26, 27], and GW dephasing $\delta\Psi(f)$, the phase shift in gravitational-wave signals detectable by LIGO/Virgo [28] or LISA [31]; these are then confronted with EHT and GW posteriors.

7 Discussion and Broader Impact

Positioning relative to other approaches. EDG can now be directly compared with other emergent gravity proposals. Verlinde’s entropic gravity [13, 14], causal set theory [15], and loop quantum gravity [16] all provide

deep conceptual avenues, but often lack falsifiable links to astrophysical data. By contrast, EDG passes Newtonian and post-Newtonian tests, respects entropy balance in line with Jacobson’s thermodynamic derivation of Einstein’s equations [3, 17], and, in our new R13p analysis, within the adopted phenomenological likelihood, EDG illustrates how entanglement-derived corrections can be explored within the computational validation framework. Within the adopted phenomenological likelihood used in the R13p analysis, model-selection metrics favor the EDG parameterization relative to the GR-limit slice of the model space, with values $\Delta\text{AIC} \approx -14$, $\Delta\text{BIC} \approx -14$, and $\Delta \log Z \approx +7$. Because these statistics depend on the simplified likelihood implementation described in Section 9, they should be interpreted as illustrative rather than definitive evidence. Their primary significance is methodological: they demonstrate that entanglement-based deformations of spacetime geometry can be quantitatively compared with astrophysical observations within a reproducible computational validation framework.

Information–geometry bridge. Earlier work connecting entanglement and geometry [5, 6, 4, 35, 7, 8, 22, 18, 20, 21] suggested that spacetime structure may arise from underlying patterns of quantum entanglement. More recent progress has further formalized this connection using tools such as entanglement wedge reconstruction, tensor networks, and information-theoretic formulations of gravity [36, 37], reinforcing the broader theoretical context in which EDG is situated.

Within the computational framework developed here, EDG provides a concrete phenomenological realization of these ideas. In the R13p analysis, the posterior concentrates near a finite entanglement scale $L_q \sim 10^{13}$ m when evaluated under the adopted phenomenological likelihood that combines Event Horizon Telescope photon-ring measurements [26, 27] with gravitational-wave phase constraints [28, 29, 30, 31]. While this result should be interpreted as an exploratory property of the implemented likelihood rather than definitive evidence for new gravitational physics, it illustrates how entanglement-based corrections to spacetime geometry can be quantitatively investigated within a reproducible computational validation framework.

Cosmology and the dark sector. The implications of an entanglement-driven modification extend to cosmology. Padmanabhan [38] and Afshordi et al. [39] argued that emergent perspectives could account for dark energy or dark matter. EDG, with its finite L_q , provides a pathway for such effects: small but cumulative entanglement corrections could mimic dark matter in galactic rotation curves or contribute to accelerated expansion. Future CMB and large-scale structure surveys can therefore serve as complementary probes alongside black-hole and GW tests.

Falsifiability and near-term tests. A critical strength

of EDG is its falsifiability. Our R13p analysis shows, within the adopted phenomenological likelihood, model selection metrics favor the EDG parameterization relative to the GR-limit slice of the model space. Yet EDG remains exposed: improvements in EHT systematics, the ngEHT, and space-based GW interferometers such as LISA and Cosmic Explorer will soon probe the finite- L_q regime at higher precision. This capacity to be confirmed or ruled out by forthcoming experiments distinguishes EDG from many speculative theories.

Interpretation of the entanglement scale. The inferred scale L_q represents an effective length characterizing the distance over which entanglement-driven corrections to the Einstein equations become relevant in the present parameterization. In the EDG framework this scale should not necessarily be interpreted as a fundamental microscopic length such as the Planck scale. Instead it may reflect a coarse-grained emergent scale arising from the collective organization of underlying quantum degrees of freedom. In information-theoretic descriptions of spacetime, entanglement entropy commonly obeys area-law behavior, suggesting that geometric quantities may encode aggregated information over extended regions. Within this perspective L_q may represent an effective correlation scale at which entanglement structure begins to influence the emergent geometry, while the parameter p controls how strongly those corrections scale with system size.

Impact. This example demonstrates how emergent spacetime models can be evaluated using reproducible computational validation pipelines. It may help unify quantum information ideas with black-hole thermodynamics [32, 33, 34] and holography [21, 20], reshaping our understanding of spacetime at a fundamental level. The transition from conceptual toy models to a data-driven alternative to GR marks EDG as a candidate for a potential avenue for connecting quantum information structures with gravitational dynamics in gravitational physics. Our results complement recent independent efforts linking gravity and quantum information [37], suggesting that the unification of entanglement, holography, and cosmology may represent the next stage in gravitational theory. Our framework resonates with recent progress on the black hole information problem. The island formula and entanglement wedge reconstruction [40, 41] have demonstrated that quantum information principles can reproduce the Page curve, reinforcing the role of entanglement in governing horizon dynamics. In this context, EDG provides a complementary route where entanglement structure not only resolves paradoxes but also quantitatively deforms spacetime geometry.

8 Reproducibility and Computational Transparency

A central objective of this work is to demonstrate that emergent spacetime hypotheses can be investigated within a reproducible computational framework rather than remaining purely conceptual proposals. The EDG implementation is therefore structured as an executable research artifact in which simulation outputs, validation stages, and observational comparisons can be regenerated deterministically from the repository.

The framework exposes clear entry points for all validation stages through the orchestration script `run_validations.py`. Intermediate outputs, including entropy diagnostics, curvature estimates, weak-field summaries, and posterior grids, are written as structured CSV artifacts that allow independent verification of the figures and tables reported in this paper. This design also allows alternative emergent gravity models to be inserted into the same validation workflow with minimal modification.

All code, parameter configurations, and intermediate data products used in this analysis are available in the accompanying repository referenced in Ref. [42]. The repository therefore functions not only as supplemental material but as a computational environment capable of reproducing the principal numerical experiments reported here.

By emphasizing executable reproducibility and modular validation, the framework provides a transparent platform for exploring information theoretic approaches to spacetime emergence and for comparing competing emergent gravity models under a common validation structure.

It is important to emphasize that the EDG framework is not intended to provide a complete microscopic theory of gravity. Rather, it serves as a computational platform for testing how information theoretic corrections to spacetime geometry might manifest in observable regimes. In this sense the framework should be viewed as a phenomenological bridge between quantum information inspired models of spacetime and the astrophysical data that can constrain them.

9 Limitations of the Current Implementation

While the EDG framework demonstrates how entanglement-derived corrections to spacetime geometry can be explored within a reproducible computational environment, the present implementation should be viewed as a phenomenological proof-of-concept rather than a complete physical theory.

First, the strong-field likelihood used in the R13p stage relies on simplified observational summaries derived from published Event Horizon Telescope and gravitational-wave measurements. The analysis therefore does not

yet incorporate the full interferometric visibility modeling used in EHT imaging pipelines or the complete Bayesian parameter-estimation frameworks employed in gravitational-wave data analysis.

Second, the informational stress tensor $X_{\mu\nu}$ is currently implemented through discretized entropy and mutual-information observables extracted from tensor-network simulations. While this approach provides a concrete computational realization of the EDG concept, it should be regarded as an effective approximation to a more fundamental microscopic description of quantum gravitational degrees of freedom.

Finally, the parameterization (ϵ, L_q, p) is intentionally minimal in order to enable interpretable inference across the multi-stage validation pipeline. Future theoretical developments may motivate richer parameterizations or alternative functional forms for entanglement-driven corrections to the Einstein equations.

For these reasons the present work should be interpreted as establishing a computational methodology for exploring information-theoretic spacetime models, rather than as providing a definitive alternative theory of gravity. The framework is designed to evolve as higher-fidelity observational likelihoods and more detailed quantum simulations become available.

10 Conclusion

We have developed and validated Entanglement-Driven Gravity (EDG), an information-theoretic framework in which spacetime geometry emerges from patterns of quantum entanglement. Through a 13-stage pipeline spanning Newtonian sanity checks, post-Newtonian observables, entropy diagnostics, convergence tests, and strong-field synthesis, EDG consistently reduces to GR where the latter is well tested, while yielding falsifiable extensions in regimes where GR remains silent.

Our strong-field analysis (R13p), combining horizon-scale photon-ring imaging from the Event Horizon Telescope with gravitational-wave inspiral phasing, identifies a posterior concentration near finite entanglement scale within the adopted likelihood. The data prefer a finite entanglement length scale $L_q \sim 10^{13}$ m, supported by robust model-selection statistics (ΔAIC , ΔBIC , $\Delta\log Z$). Within the adopted phenomenological likelihood, the analysis indicates that a finite entanglement scale L_q is statistically preferred relative to the GR-limit slice of the parameterization. This result illustrates how observational data can be used to explore entanglement-based extensions of gravity within a reproducible computational framework.

Looking forward, EDG is directly testable: near-term facilities (ngEHT, LISA, Cosmic Explorer) will probe the finite- L_q regime at higher precision, with the power to confirm or falsify its predictions. Beyond astrophysics, cumulative entanglement corrections may extend to cos-

mology, providing a pathway to address dark matter and dark energy within an information-theoretic framework.

In sum, EDG unites quantum information theory, black-hole thermodynamics, and astrophysical inference under a single, data-driven paradigm. By linking entanglement gradients to curvature sources, it offers a constructive bridge between microscopic quantum correlations and macroscopic geometry. If borne out, EDG would signal a paradigm shift: gravity not only geometric but informational at its core, with the fabric of spacetime woven from entanglement itself.

Key Takeaways:

- EDG recovers GR in all tested Newtonian and post-Newtonian regimes, validating its consistency with established physics.
- Strong-field analysis (R13p) combining EHT and GW data identifies a posterior concentration near finite entanglement scale within the adopted likelihood, with finite $L_q \sim 10^{13}$ m preferred.
- EDG is falsifiable: next-generation facilities (ngEHT, LISA, Cosmic Explorer) will confirm or rule out its predictions in the near term.
- This example demonstrates how emergent spacetime models can be evaluated using reproducible computational validation pipelines.

Limitations and Future Work

The strong-field validation stage (R13p) currently employs phenomenological likelihood models constructed from published observational summaries. While this approach allows exploratory testing of emergent spacetime models, it does not yet incorporate full observational pipelines such as interferometric visibility modeling for EHT data or full gravitational-wave parameter estimation frameworks.

Future work will extend the framework to incorporate higher-fidelity observational likelihoods and more detailed astrophysical modeling. Additional directions include exploring alternative entanglement-based gravity models, improving tensor-network simulation fidelity, and integrating large-scale structure and cosmological observables.

Closing Statement. Entanglement-Driven Gravity (EDG) moves beyond pure speculation in a computational sense: it is implemented as a falsifiable, reproducible case study that unites quantum information, black-hole thermodynamics, and astrophysical inference within a single executable framework.

Appendix: Extended Methods

Implementation of the EDG informational-stress equation. The novel EDG framework is implemented in the codebase not as a single monolithic routine but as a distributed set of modules that together realize an informational-stress source term. The entanglement density ρ_{ent} and stress-energy analogue $T_{\mu\nu}^{(\text{ent})}$ are computed in `einstein_tensor.py`, which approximates an Einstein-like tensor from mutual information and entropy inputs. Entropy gradients (∇S) are evaluated in `entropy.py`, while curvature and Laplacian-like contributions ($\nabla^2 S$) are assembled in `curvature.py`. These are injected into the potential/force update in `solar_system_entanglement.py`, where the effective Newtonian pull is modulated by a blend of dS/dt , Hawking-like mutual information, and optional spatial curvature multipliers from the lattice. The orchestration of this pipeline is handled by `entanglement_spacetime_evolution.py`, which evolves the PEPS tensor network, computes the informational sources, and exports CSVs (`entropy.csv`, `hawking_radiation.csv`, `curvature_lattice.csv`) that feed into downstream validations and visualizations. Thus the informational-stress equation is concretely realized in the framework: ρ_{ent} acts as the source density and $T_{\mu\nu}^{(\text{ent})}$ provides the stress tensor, both coupled discretely to potential updates in the orbital and black-hole demos.

For the strong-field R13p stage, the joint likelihood implements fractional EHT ring-size shifts and a toy gravitational-wave phase correction consistent with the same (L_q, p) scaling. While not exposed as a named additive $\Psi_{\text{GR}} + \delta\Psi$ function, this correction is included implicitly in the joint observational fit modules. The 13-stage pipeline (`run_validations.py`) therefore operates on the phenomenological parameterization (ϵ, L_q, p) , while the quantum simulation side computes S_{ent} , mutual information, and curvature sources that prototype the full solver.

EHT data handling (reproducibility note). The R13p script does not read a bundled EHT data file. Instead, it constructs default EHT inputs in-code from published values (e.g., Sgr A*: $51.8 \pm 3.0 \mu\text{as}$; M87*: $42 \pm 3 \mu\text{as}$), and computes the Schwarzschild radius $r_s = 2GM/c^2$ from the supplied or default black-hole mass. These defaults can be overridden either via command-line arguments (e.g., `--add_eht sgrA` or `--add_eht m87` with `--eht_mode {fractional, absolute}`, `--eht_mass`, `--eht_DGR`, `--eht_sigma_abs`, `--eht_mu_frac`, `--eht_sigma_frac`) or by providing an external CSV via `--ext_csv`. The script then forms the joint likelihood with GW rows and writes all derived products (posterior grid, 1D marginals, constraints tables, and Fig. 7). For full transparency, the defaults and overrides are doc-

umented in `entanglement_validation/validation_r13p_joint_observational_fit.py`.

Numerical integrators. Newtonian and post-Newtonian trajectories were integrated using high-precision RK4 and symplectic Verlet methods, with step-size refinement ladders to verify convergence. Energy drift was monitored over 10^5 steps to ensure stability at the 10^{-12} level.

Bootstrap and consensus. Weak-field parameter inferences were tested with bootstrap resampling (~ 500 draws per integrator) with convergence monitored by effective sample size (ESS) and Gelman–Rubin statistics, following standard statistical practice [43]. Complementary ϵ sweeps probed sensitivity of the perihelion parameter A_0 . Cross-integrator comparisons confirmed consensus between RK4 and Verlet baselines.

Joint likelihood construction (R13p). For the strong-field analysis, we constructed a joint likelihood combining: (i) Event Horizon Telescope (EHT) ring-size measurements, modeled as fractional shifts $\Delta d/d_{\text{GR}} \sim \kappa(L_q/r_s)^p$ relative to GR expectations, and (ii) gravitational-wave inspiral phase corrections $\delta\Psi(f)$ derived from entanglement-driven deformations of the post-Newtonian expansion. Weak-field priors from R4 and R7–R9 were imposed.

Posterior sampling. The (ϵ, L_q, p) parameter space was discretized on a three-dimensional grid spanning $\epsilon \in [0.9, 1.1]$, $\log_{10}(L_q/\text{m}) \in [12, 14]$, and $p \in [0, 4]$. Marginals and joint constraints were obtained by summing the likelihood over this grid. Convergence was cross-checked by varying resolution and comparing interpolated posteriors.

Model comparison. Model-selection statistics were computed to directly compare EDG against GR (the $\epsilon = 1$, $L_q \rightarrow \infty$ limit). Information criteria were evaluated as $\text{AIC} = 2k - 2 \ln \mathcal{L}_{\text{max}}$ and $\text{BIC} = k \ln N - 2 \ln \mathcal{L}_{\text{max}}$, where k is the number of parameters and N the number of data points. Bayesian evidence Z was estimated by trapezoidal integration over the posterior grid. The resulting $\Delta\text{AIC} \approx -14$, $\Delta\text{BIC} \approx -14$, and $\Delta \log Z \approx +7$ should be interpreted as properties of the implemented likelihood comparison, illustrating how entanglement-derived corrections can be explored within the computational validation framework.

Computational scaling. Strong-field likelihood evaluations required $\mathcal{O}(10^6)$ orbit and waveform samples. These were distributed across parallel CPU nodes, and the analyses completed within routine wall-clock times on modest resources. The framework scales efficiently and remains compatible with higher-resolution analyses and forecasts for next-generation detectors.

Relation to discrete geometry approaches. The discrete-curvature implementation in EDG resonates with causal dynamical triangulations [44, 45] and Regge calculus [46], where spacetime geometry is approximated via simplicial or lattice methods. In our case, entanglement-derived informational sources ($\rho_{\text{ent}}, T_{\mu\nu}^{\text{ent}}$) couple directly to curvature computed from tensor-network lattices, bridging information-theoretic inputs with discrete gravitational analogues.

Data availability and executable artifact. Data supporting the results of this study are available through the repository and accompanying CSV outputs used throughout the validation stages. The uploaded files include weak-field summaries, sensitivity sweeps, and the R13p posterior grid used to build the figures in this paper. Together with the orchestration script `run_validations.py`, these artifacts make it possible to regenerate the principal tables and plots reported here, supporting the paper’s computational-science emphasis on executable reproducibility.

Data availability. Data supporting the results of this study are available within the article and its Supplemental Material. The Supplemental Material includes CSV files (e.g., ‘entropy.csv’, ‘curvature_lattice.csv’, ‘R13p_posterior_grid.csv’) containing raw data from the 13-stage validation pipeline, including entropy diagnostics, curvature evolution, and the R13p joint fit posterior. Additional data and processing code are accessible at the GitHub repository [42], where scripts (e.g., ‘validation_r13p_joint_observational_fit.py’) and further CSV outputs are provided.

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References

- [1] Jacob D. Bekenstein. Black holes and entropy. *Physical Review D*, 7(8):2333–2346, 1973.
- [2] S. W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics*, 43:199–220, 1975.
- [3] Ted Jacobson. Thermodynamics of spacetime: The einstein equation of state. *Physical Review Letters*, 75(7):1260–1263, 1995.
- [4] S. Ryu and T. Takayanagi. Holographic derivation of entanglement entropy from the ads/cft correspondence. *Physical Review Letters*, 96:181602, 2006.
- [5] Mark Van Raamsdonk. Building up spacetime with quantum entanglement. *General Relativity and Gravitation*, 42:2323–2329, 2010.
- [6] H. Casini, M. Huerta, and R. C. Myers. Towards a derivation of holographic entanglement entropy. *JHEP*, 2011(05):036, 2011.
- [7] Juan Maldacena and Leonard Susskind. Cool horizons for entangled black holes. *Fortschritte der Physik*, 61:781–811, 2013.
- [8] Leonard Susskind. Er=epr, ghz, and the consistency of quantum measurements. *Fortschritte der Physik*, 64(1):49–71, 2016.
- [9] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. SIAM, 2 edition, 2002.
- [10] Victoria Stodden, Jennifer Seiler, and Zhaokun Ma. An empirical analysis of journal policy effectiveness for computational reproducibility. *Proceedings of the National Academy of Sciences*, 115(11):2584–2589, 2018.
- [11] M. E. J. Newman. *Networks*. Oxford University Press, 2 edition, 2018.
- [12] Román Orús. Tensor networks for complex quantum systems. *Nature Reviews Physics*, 1:538–550, 2019.
- [13] Erik Verlinde. On the origin of gravity and the laws of newton. *Journal of High Energy Physics*, 2011(4):29, 2011.
- [14] Erik Verlinde. Emergent gravity and the dark universe. *SciPost Physics*, 2:016, 2017.
- [15] Rafael D. Sorkin. Causal sets: Discrete gravity. In A. Gomberoff and D. Marolf, editors, *Lectures on Quantum Gravity*, pages 305–327. Springer, 2005.
- [16] Carlo Rovelli and Lee Smolin. Discreteness of area and volume in quantum gravity. *Nuclear Physics B*, 442:593–619, 1995.
- [17] Ted Jacobson. Entanglement equilibrium and the einstein equation. *Physical Review Letters*, 116(20):201101, 2016.
- [18] Brian Swingle. Constructing holographic spacetimes using entanglement renormalization. *Annals of Physics*, 339:98–114, 2013.
- [19] G. Vidal. Entanglement renormalization. *Physical Review Letters*, 99:220405, 2007.
- [20] Glen Evenbly and Guifre Vidal. Tensor network renormalization. *Physical Review Letters*, 115(18):180405, 2015.

- [21] Fernando Pastawski, Beni Yoshida, Daniel Harlow, and John Preskill. Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence. *Journal of High Energy Physics*, 2015(6):149, 2015.
- [22] V. Balasubramanian, B. Czech, B. D. Chowdhury, J. de Boer, and M. P. Heller. Entwinement and the emergence of spacetime. *JHEP*, 2015(01):048, 2015.
- [23] Clifford M. Will. The confrontation between general relativity and experiment. *Living Reviews in Relativity*, 17(1):4, 2014.
- [24] Irwin I. Shapiro. Fourth test of general relativity. *Physical Review Letters*, 13(26):789–791, 1964.
- [25] Bruno Bertotti, Luca Iess, and Paolo Tortora. A test of general relativity using radio links with the cassini spacecraft. *Nature*, 425(6956):374–376, 2003.
- [26] The Event Horizon Telescope Collaboration. First m87 event horizon telescope results. i. the shadow of the supermassive black hole. *The Astrophysical Journal Letters*, 875(1):L1, 2019.
- [27] Event Horizon Telescope Collaboration. First m87 event horizon telescope results. vi. the shadow and mass of the central black hole. *ApJL*, 875(1):L6, 2019.
- [28] B. P. Abbott et al. Tests of general relativity with gw150914. *Physical Review Letters*, 116(22):221101, 2016.
- [29] Nicolas Yunes and Xavier Siemens. Gravitational-wave tests of general relativity with ground-based detectors and pulsar-timing arrays. *Living Reviews in Relativity*, 16(1):9, 2013.
- [30] Emanuele Berti, Vitor Cardoso, Andrea Maselli, Leonardo Gualtieri, and Henrique W. Hammer. Testing general relativity with present and future astrophysical observations. *Classical and Quantum Gravity*, 32(24):243001, 2015.
- [31] LISA Consortium. Laser interferometer space antenna, 2017.
- [32] Don N. Page. Information loss in black holes. *Physical Review Letters*, 71(23):3743–3746, 1993.
- [33] A. Almheiri, D. Marolf, J. Polchinski, and J. Sully. Black holes: complementarity or firewalls? *JHEP*, 2013(02):062, 2013.
- [34] Sasha Haco, Stephen W. Hawking, Malcolm J. Perry, and Andrew Strominger. Black hole entropy and soft hair. *Journal of High Energy Physics*, 2018(12):098, 2018.
- [35] T. Faulkner, A. Lewkowycz, and J. Maldacena. Quantum corrections to holographic entanglement entropy. *JHEP*, 2013(11):074, 2013.
- [36] Ning Bao, Ven Chandrasekaran, and Geoff Penington. Gravity from quantum information. *Journal of High Energy Physics*, 2022(1):114, 2022.
- [37] Shadi Ali Ahmad, Stefano Bolognesi, Luca Ciambelli, and Luciano Vanzo. Gravity from quantum information: The emergence of spacetime geometry. *Annals of Physics*, 472:169428, 2025.
- [38] T. Padmanabhan. Emergent perspective of gravity and dark energy. *Research in Astronomy and Astrophysics*, 12(8):891–916, 2012.
- [39] Niayesh Afshordi et al. Emergent gravity and the dark universe. *Physical Review D*, 94(8):084044, 2016.
- [40] Geoff Penington. Entanglement wedge reconstruction and the information paradox. *Journal of High Energy Physics*, 2020(9):002, 2020.
- [41] Ahmed Almheiri, Raghu Mahajan, Juan Maldacena, and Ying Zhao. The page curve of hawking radiation from semiclassical gravity. *Journal of High Energy Physics*, 2020(3):149, 2020.
- [42] Kenneth G. II Young. Entanglements spacetime: Edg validation pipeline and analysis code, 2025. Accessed: September 15, 2025.
- [43] Bradley Efron and Robert Tibshirani. *An Introduction to the Bootstrap*, volume 57 of *Monographs on Statistics and Applied Probability*. CRC Press, 1994.
- [44] J. Ambjørn, J. Jurkiewicz, and R. Loll. Reconstructing the universe. *Phys. Rev. D*, 72:064014, 2005.
- [45] J. Ambjørn, J. Jurkiewicz, and R. Loll. The universe from scratch. *Contemporary Physics*, 47(2):103–117, 2006.
- [46] T. Regge. General relativity without coordinates. *Nuovo Cimento*, 19:558–571, 1961.